THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A ball dropped from 5 m rebounds to 3.2 m. The fraction of energy lost is: 1

(a) 50 per cent **(b)** 64 per cent **(c)** 36 per cent **(d)** 3.2 per cent

2. The loudness of sound is measured in: 1

(a) watts **(b)** hertz **(c)** metres **(d)** decibels

3. The displacement of a body after half a circle of radius r is: 1

(a) r **(b)** 2r **(c)** pi r **(d)** zero

4. A man pushes a box of 20 kg with 50 N and the box moves at constant velocity. The friction is: 1

(a) 200 N **(b)** 50 N **(c)** 20 N **(d)** 0 N

5. The kinetic energy of a body is 50 J and its mass is 4 kg. The velocity is: 1

(a) 25 m/s **(b)** 3.5 m/s **(c)** 5 m/s **(d)** 12.5 m/s

6. A tuning fork of 512 Hz produces sound of wavelength 0.67 m. The speed is about: 1

(a) 0.0013 m/s **(b)** 512.67 m/s **(c)** 756 m/s **(d)** 343 m/s

7. A car moving at 72 km/h stops in 8 s. The retardation is: 1

(a) 9 m/s squared **(b)** 0.9 m/s squared **(c)** 4 m/s squared **(d)** 2.5 m/s squared

8. A force acts on a mass of 2 kg for 4 s starting from rest and gives it a velocity of 8 m/s. The force is: 1

(a) 4 N **(b)** 8 N **(c)** 16 N **(d)** 2 N

9. A motor lifts 100 kg to 15 m in 30 s. The power is (g = 10): 1

(a) 50 W **(b)** 500 W **(c)** 5000 W **(d)** 1500 W

10. If the frequency is doubled, the pitch: 1

(a) becomes lower **(b)** stays same **(c)** becomes zero **(d)** becomes higher

11. In a v-t graph, a line sloping downward from 20 m/s to 0 in 4 s shows retardation of: 1

(a) 4 m/s squared **(b)** 80 m/s squared **(c)** 5 m/s squared **(d)** 20 m/s squared

12. A lift accelerates upward at 2 m/s squared. A person of mass 50 kg in it feels a force from the floor of: 1

(a) 600 N **(b)** 400 N **(c)** 100 N **(d)** 500 N

13. A load of 150 N is raised by an effort of 50 N by a machine. The MA is: 1

(a) 3 **(b)** 1/3 **(c)** 200 **(d)** 100

14. The time between two consecutive compressions at a point is the: 1

(a) time period **(b)** amplitude **(c)** wavelength **(d)** speed

15. The distance covered by a body after half a circle of radius r is: 1

(a) pi r **(b)** 2r **(c)** 2 pi r **(d)** zero

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Calculate the MA of a lever with load arm 25 cm and effort arm 1 m and the effort required for a 400 N load. 2

17. How do you get the speed of sound with a gun and a stopwatch method? Describe in two points. 2

18. A scooter changes its velocity from 6 m/s to 12 m/s in 3 s. Find the acceleration and the distance moved. 2

19. Find the weight of a 70 kg person on the Moon (g_moon = 1.7 m/s squared) and compare with that on Earth (g = 10). 2

20. A man lifts a load of 25 kg to 2 m in 5 s. Find the work and the power (g = 10). 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A crane lifts a 500 kg load through 20 m in 40 s. Find the work done and power (g = 10). 3

22. A ship sends an ultrasonic signal and receives the echo from the sea bed 2 s later. Find the depth (v = 1500 m/s) and the time if the depth were doubled. 3

23. The speed-time data of a car are: t(s) 0, 2, 4, 6 and v(m/s) 0, 6, 12, 18. Find the acceleration and the distance in 6 s. 3

24. Explain why it is easier to pull a lawn roller than to push it. Use the idea of friction and the vertical component of force. 3

OR

Explain the use of an inclined plane with a calculation: a 300 N drum is raised by 1 m using a 5 m ramp. Find effort (ideal) and MA. 3

25. A bat emits an ultrasonic pulse of 50 kHz. Find the wavelength in air (v = 340 m/s) and the time for the echo from an insect 1.7 m away. 3

OR

Show that for uniform acceleration the average velocity equals (u + v)/2 and use it to derive s = (u + v)t/2. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A ball is thrown vertically upward from the ground with 30 m/s. Find 5

(a) the time to reach the top

(b) the maximum height

(c) the total time of flight

(d) the velocity on returning (g = 10).

OR

A gun of mass 4 kg fires a bullet of mass 20 g with a velocity of 400 m/s. Find the momentum of the bullet and the recoil velocity of the gun using equal and opposite momenta. 5

27. Explain the working of a hydroelectric power plant and calculate the power available if 5000 kg of water falls per second through 40 m with 80 per cent efficiency (g = 10). 5

OR

Explain the human ear and how the signal travels from the outer ear to the brain. State two precautions to protect hearing. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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